



Nebraska Public Power District

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NLS2019041
October 7, 2019

U.S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D.C. 20555-0001

Subject: Licensee Event Report No. 2019-002-00
Cooper Nuclear Station, Docket No. 50-298, DPR-46

Dear Sir or Madam:

The purpose of this correspondence is to forward Licensee Event Report 2019-002-00.

There are no new commitments contained in this letter.

Sincerely,

John Dent, Jr.
Vice President Nuclear -
Chief Nuclear Officer

/jo

Attachment: Licensee Event Report 2019-002-00

cc: Regional Administrator w/attachment
USNRC - Region IV

NPG Distribution w/attachment

Cooper Project Manager w/attachment
USNRC - NRR Plant Licensing Branch IV

INPO Records Center w/attachment
via ICES entry

Senior Resident Inspector w/attachment
USNRC - CNS

SORC Chairman w/attachment

SRAB Administrator w/attachment

CNS Records w/attachment

IEZZ
NRR



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form <http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name
Cooper Nuclear Station

2. Docket Number
05000298

3. Page
1 of 4

4. Title

Secondary Containment Rise in Differential Pressure Results in Loss of Safety Function

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
08	08	2019	2019	- 002	- 00	10	07	2019	Facility Name	05000
									Facility Name	Docket Number
										05000

9. Operating Mode

1

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)

10. Power Level

100

☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
☐ 50.73(a)(2)(i)(C)		☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact

Linda Dewhirst, Regulatory Affairs and Compliance Manager

Telephone Number (Include Area Code)

(402) 825-5416

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To ICES	Cause	System	Component	Manufacturer	Reportable To ICES
A	NG	FLT		Y					

14. Supplemental Report Expected

☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No

15. Expected Submission Date

Month	Day	Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On August 8, 2019, a spike in Secondary Containment occurred with differential pressure (DP) exceeding the Technical Specification limit of 0.25 inch of vacuum water gauge (wg) causing Secondary Containment to be inoperable for 7 minutes. The spike was due to Reactor Building heating, ventilation, and air conditioning (HVAC) supply filter media being advanced excessively causing DP to rise to 0.19 vacuum wg. The DP was not immediately identified by Control Room personnel due to Secondary Containment DP not reaching the high pressure alarm setpoint and they were unaware of the filters being advanced.

Causal factors were that the condition of the supply filters was not properly prioritized and risk associated with the supply filter activities was not recognized and controlled properly due to it being classified as a skill of the craft activity.

The operations department skill of the craft list was revised to delete activities such as advancing the supply filters that have plant operational risk. More formal controls are being implemented for operation and maintenance of the reactor building HVAC supply filters. Operations department wide communications are planned to discuss the event, causal factors, and corrective actions. There was no impact on nuclear safety, plant reliability, radiological safety, or industrial safety.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME		2. DOCKET NUMBER		3. LER NUMBER		
Cooper Nuclear Station		05000298		YEAR	SEQUENTIAL NUMBER	REV NO.
				2019	- 002	- 00

NARRATIVE**PLANT STATUS**

Cooper Nuclear Station was in Mode 1, Power Operation, at 100 percent power, when the event was discovered; i.e., August 8, 2019.

BACKGROUND

Technical Specification (TS) Surveillance Requirement 3.6.4.1.1 verifies that Secondary Containment vacuum is greater than or equal to 0.25 inch of vacuum water gauge (wg). Limiting Condition of Operation (LCO) 3.6.4.1, requires Secondary Containment to be operable in Mode 1. If Secondary Containment is not operable, LCO 3.4.6.1, Condition A, requires restoration of Secondary Containment to operable within 4 hours.

The Secondary Containment system is designed with sufficient redundancy so that no single active system component failure can prevent the system from achieving its safety objectives, which includes preventing ground level releases of airborne radioactive materials to the environment. Secondary Containment consists of four main subsystems, two of these are the Reactor Building [EIS: NG] and standby gas treatment system [EIS: BH]. The Secondary Containment system is designed to be sufficiently leak tight to allow the standby gas treatment system to reduce the Reactor Building pressure to a minimum of 0.25 inch of water column vacuum when neutral wind conditions exist when the standby gas treatment system fans are exhausting Reactor Building atmosphere at a rate of 100% per day of the Reactor Building free volume.

During normal operations, the Reactor Building ventilation system [EIS: VA] is operated to produce a minimum of 0.25 inch of water column vacuum inside the Reactor Building as compared to the outside atmosphere.

The Reactor Building supply filter consists of three separate rolls of filter media, each one being five feet wide. The new filter rolls load onto the filter unit at the top of the unit and the filter media is pulled out and connected to take up reels at the bottom of the unit. The bottom rolls are connected to motors that are operated by a switch. When the switch is activated, all three motors actuate and the three bottom take up spools pull the filter media down across the screen face to expose fresh filter media to the supply air flow. The filter rolls include paper leaders and trailers to assist in installation.

EVENT DESCRIPTION

In May 2019, the south roll of filter media was becoming depleted and needed replaced; however there was no replacement fan filter media on-site. This issue was not added to the short-term operational issues list and filter replacement was not appropriately prioritized, resulting in a delay of approximately three months to receive the filter media.

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Cooper Nuclear Station	05000298	2019	- 002	- 00

On August 8, 2019, at 1025, during a panel walkdown, it was discovered that a spike to 0.19 vacuum wg in Secondary Containment had occurred which exceeded the TS limit of 0.25 inch of vacuum wg between the time of 0848 to 0855, causing Secondary Containment to be inoperable for 7 minutes.

Investigation revealed that the cause of the spike was due to the Reactor Building heating, ventilation, and air conditioning (HVAC) supply filter media being advanced excessively. The Reactor Building Station Operator found the filter media excessively dirty and advanced the filter roll to restore the filters to a clean condition. This caused the Reactor Building HVAC flow to rise and Reactor Building differential pressure (DP) exceeded 0.25 inch of vacuum wg. The DP excursion was not immediately identified by Control Room personnel as they were not aware that the filters were being advanced and Secondary Containment DP did not reach the high pressure alarm setpoint.

BASIS FOR REPORT

This event is reportable as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

An engineering review was performed and the results demonstrate the perturbation on Secondary Containment pressure caused by the non-essential Reactor Building HVAC system did not impact the integrity of Secondary Containment and had no impact on the Standby Gas Treatment system. In the event of an accident, the Standby Gas Treatment system would have automatically started and would have been able to maintain vacuum on Secondary Containment as assumed in the Loss of Coolant Accident dose analysis. As such, this event will not be counted as a Safety System Functional Failure for the Nuclear Regulatory Commission performance indicator since no loss of safety function occurred.

SAFETY SIGNIFICANCE

The pressure excursion in the Reactor Building resulted in an unexpected plant condition and met the entry condition for required action pursuant to Technical Specifications. Control Room operators were challenged to promptly identify the Secondary Containment high DP condition because the Reactor Building high DP alarm setpoint was not reached during the event. This event resulted in a minor excursion of Reactor Building DP above the TS limit of 0.25 inches vacuum wg for several minutes until the building HVAC controls recovered DP. This event did not negate the ability of the Standby Gas Treatment System to establish a Secondary Containment negative pressure. There was no impact on nuclear safety, plant reliability, radiological safety, or industrial safety.

CAUSE

Two causal factors were identified. The first being that the stations short-term issue list was not utilized to properly prioritize and address the condition of the Reactor Building HVAC supply filters. The second is that the risk associated with Reactor Building HVAC supply filter activities was not recognized and controlled properly due to it being classified as a skill of the craft activity.

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Cooper Nuclear Station	05000298	YEAR	SEQUENTIAL NUMBER	REV NO.
		2019	- 002	- 00

CORRECTIVE ACTIONS

The Operations department skill of the craft list was revised to delete activities, such as advancing Reactor Building HVAC supply filters, that have plant operational risk. More formal controls are being implemented for operation and maintenance of the Reactor Building HVAC supply filters. Operations department wide communications are planned to discuss this event, causal factors, and corrective actions.

PREVIOUS EVENTS

On January 6, 2014, Secondary Containment was declared inoperable due to a rise in differential pressure. This was reported under LER 2014-001-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.

On September 10, 2012, Secondary Containment was breached due to both airlock doors in the Reactor Building being inadvertently left open simultaneously. This was reported under LER 2012-003-00 as a loss of safety function under 10 CFR 50.73(a)(2)(v)(C and D) – An event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material or mitigate the consequences of an accident.